INCH-POUND

MIL-N-53093(ME)
7 February 1990

MILITARY SPECIFICATION

NOZZLE, ADAPTER, CLOSED-CIRCUIT TO GRAVITY FILL,

ARCTIC SERVICE

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a nonautomatic shutoff adapter nozzle, intended for adapting a closed-circuit refueling nozzle (conforming to MIL-N-53094) to perform conventional open port or gravity fill dispensing of liquid hydrocarbon fuels in a low temperature (-60 °F) environment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-P-416	- Plating, Cadmium (Electrodeposited).
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-C-1752	- Cushioning Material, Packaging, (Uncellular Polyethylene Foam, Flexible).
PPP-C-1842	- Cushioning Material, Plastic, Open Cell (For Packaging Applications).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4930

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MIL-P-116	- Preservation, Methods of.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-C-7024	- Calibrating Fluid, Aircraft Fuel System Components.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-N-53094	- Nozzle Assembly, Closed-Circuit Refueling, Arctic Service.
MIL-C-81562	- Coating, Cadmium, Tin Cadmium and Zinc (Mechanically Deposited).
MIL-C-81740	- Coating, Aluminum and Aluminum Alloys (Metallic Compound Decomposition).
MIL-C-81797	- Coating, Inorganically Bonded Aluminum (Electrophoretically Deposited).
MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.
FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-870	- Cadmium Plated, Low Embrittlement, Electrodeposition.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1	- Surface Texture.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 276	- Stainless and Heat-Resisting Steel Bars and Shapes.
A 313	- Chromium-Nickel Stainless and Heat-Resisting Steel Spring Wire.
A 380	- Cleaning and Discaling Stainless Steel Part, Equipment and Systems.
A 525	- Steel Sheet, Zinc-Coated (Galvanized) by the Hot-Dip Process, General Requirements for.
A 582	- Free-Machining Stainless and Heat - Resisting Steel Bars, Hot-Rolled or Cold-Finished.
B 26	- Aluminum-Alloy Sand Castings.
B 159	- Phosphor Bronze Wire.
B 210	- Aluminum Alloy Drawn Seamless Tubes.
D 471	- Rubber Properrty - Effects of Liquids.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

UNDERWRITERS' LABORATORIES (UL)

UL 842	- Standard for Safety: Valves for Flammable Fluids.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, LI, NY 11747; 1655 Scott Boulevard, Santa Clara, CA 95050; or 2602 Tampa East Boulevard, Tampa, FL 33619.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The close-circuit to gravity fill adapter nozzle, hereinafter referred to as a "nozzle", shall be a device for connection to a closed-circuit fuel servicing nozzle conforming to MIL-N-53094 to allow conventional refueling through an open port. The nozzle as shown in figure 1 shall consist of an inlet fitting, a main body containing a manually operated flow control valve, control lever with a protective guard, a spout, and dust caps for the inlet fitting and spout.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Materials shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

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3.3.1 Material deterioration prevention and control. The nozzles shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the nozzles may be exposed. Fuel wetted surfaces shall be compatible with all military hydrocarbon fuels. The use of copper, copper alloys or cadmium platings shall be discouraged.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2. Recovered materials. For the purpose of this requirement, recovered materials are those materials that have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the nozzles may be newly fabricated from recovered materials to the maximum extent practicable, provided the nozzles produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the nozzles.

3.3.3 Body. The nozzle body shall be cast aluminum alloy conforming to ASTM B 26, alloy UNS A03560, temper T6; alloy UNS A07120, temper T5; or alloy UNS A05350, temper F.

3.3.4 Valve springs. The valve springs shall be copper wire conforming to ASTM B 159, alloy UNS C51000, corrosion-resisting steel wire conforming to ASTM A 313, UNS S30200, S30400, or S31600, or spring steel plated in accordance with QQ-P-416, class 1, type II. In addition the spring steel can be coated according to MIL-STD-870, class 1, type I, or aluminum coated in accordance with MIL-C-81740, class 1, type II, or equivalent, or coated in accordance with MIL-C-81562, class 1, type I; or a metallic ceramic coating (0.001-inch thickness) in accordance with MIL-C-81797. Alternatively, a corrosion-resisting metal spring material with equivalent or greater corrosion resistance than the materials specified, may be used, provided the springs are capable of the same cyclic performance required for Underwriters' Laboratories, Inc., approved fuel servicing nozzles.

3.3.5 Valve stem. The valve stem shall be corrosion-resisting steel conforming to ASTM A 276, UNS S30200, S30400, S31600, or ASTM A 582, UNS S30300, condition optional, cold finish and passivated in accordance with ASTM A 380.

3.3.6 Packing. The valve stem packing shall be a solid, machined, or preformed tetrafluoroethylene polymer or a low temperature elastomer.

3.3.7 Spout. The spout shall be seamless aluminum tubing conforming to ASTM B 210, alloy A96061, temper T6, or a high strength aluminum alloy with equivalent or greater tensile strength and corrosion resistance. The spout shall

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be straight and shall be welded or threaded to the valve body. The spout shall be not less than 7.0 inches in length and not exceed 1.77 inches outside diameter.

3.3.8 Inlet fitting. The inlet fitting shall be a quick-disconnect, self-sealing automatic shutoff type connector conforming to the design and dimensions shown on figure 2.

3.4 Construction. The nozzle shall meet the performance requirements specified in 3.5, as applicable, and shall withstand the strains, jars, vibrations, and other conditions incidental to shipping, storage, and service usage. The spout centerline shall be 135 degrees from the inlet fitting centerline.

3.4.1 Weight. The nozzle shall weigh not more than 2.75 pounds.

3.4.2 Flow control. The operator shall initiate and maintain flow through the nozzle valve while utilizing a minimum operator activated force. Engagement and operation of the nozzle valve shall not be by squeezing an operating lever. The nozzle flow shall not be through a handle. Hold-open notches, lugs or lever support shall not be used.

3.4.3 Nozzle arctic operation. The nozzle shall allow for handling and operation using an arctic mitten. Any handle that shall be used with an arctic mitten shall require a minimum hand width of 5.25 inches and finger height of 3.0 inches when gripping the handle.

3.4.4 Threads. All pipe and machine screw threads shall conform to FED-STD-H28.

3.4.5 Valve stem. The valve stem surface that will come in contact with the packing during operation of the nozzle shall have a surface roughness not greater than 16 microinches. The surface roughness shall be determined in accordance with ANSI B46.1.

3.4.6 Interchangeability. All components of any one nozzle shall be interchangeable with the like part of any other nozzle furnished under any one contract.

3.4.7 Dust cap. Each nozzle shall be equipped with close fitting aluminum alloy dust caps for the spout and inlet fitting. The caps shall be of a standard design with dimensions as required to adequately protect the spout and inlet fitting. The caps shall be attached to the nozzle by means of a coated wire of aluminum or stainless steel or size no. 8 sash chain of aluminum, stainless steel, galvanized steel as per ASTM A 525 to a G90-coating designation, or brass. The connection between the cap and chain or wire shall be made using a swivel and attachment to the nozzle shall be by an S-hook. The wire or chain shall have a working tensile strength of not less than 70 pounds. A phosphor bronze spring shall be provided to maintain the caps in place over the nozzle inlet and spout tip. The spring shall be attached to the wire or to any two links of the chain in such a manner that would prevent the extension of the spring beyond its elastic limit, yet allow the caps to be snapped in place or removed from inlet and spout tip.

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3.4.8 Human factors engineering. The nozzle shall conform to human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited, to general requirements (4), design for maintainability (5.9), and hazards and safety (5.13) of MIL-STD-1472, as applicable.

3.5 Performance characteristics.

3.5.1 Hydrostatic strength. All parts of the nozzle which are subjected to pressure during normal operation shall be capable of withstanding, without rupture or any deformation, a hydrostatic pressure of 150 pounds per square inch gauge (psig).

3.5.2 Leakage. When tested as specified in 4.5.2.3, there shall be no leakage past the valve seat or through the body of any of the joints or seals of the nozzle.

3.5.3 Mechanical strength. When tested as specified in 4.5.2.4, the nozzle shall show no evidence of damage that could result in malfunction or leakage.

3.5.4 Capacity. The nozzle shall have a maximum capacity of not less than 50 gallons per minute (gpm) when tested in accordance with 4.5.2.5.

3.6 Marking for identification. Each nozzle shall be permanently and legibly marked with the manufacturer's identification by rolling, stamping, or stenciling in accordance with MIL-STD-130.

3.7 Finish. The aluminum components of each nozzle shall be anodized. The spout shall be anodized to conform to MIL-A-8625, type III, to a thickness of 0.0007-inch minimum. All other aluminum components of the nozzle shall be anodized to conform to MIL-A-8625, type II, class 2 to a thickness of 0.0007-inch minimum. The anodic coating of the visible components which require type II, class 2 coating shall be dyed green, color no. 34094 of FED-STD-595.

3.8 Maintainability. The nozzle shall operate as specified herein without the need for scheduled servicing or adjustment.

3.9 Lubrication. The nozzle shall operate satisfactorily without the use of lubricants.

3.10 Environmental.

3.10.1 Operating conditions. The nozzle shall perform as specified herein during and after exposure to the following environments:

a. Temperature range: -60 to +95 °F.
b. Relative humidity: Zero to 100 percent.
c. Salt spray: Exposure to salt sea atmosphere.

3.10.2 Storage. The nozzle shall not be damaged by storage at all ambient temperatures between -65 and +145 °F.

3.11 Workmanship. The finished nozzle shall be internally and externally clean and free of sharp edges, burrs, cracks, chips and gouges.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Disassembly inspection. After all tests, each first article nozzle shall be disassembled in the presence of a Government representative. Each disassembled part shall be examined for compliance with this specification in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected.

4.1.3 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.7).
d. Inspection comparison (see 4.6).

4.3 First article inspection.

4.3.1 Examination. Nozzles shall be examined as specified in 4.5.1. The presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Each nozzle shall be tested as specified in 4.5.2.1 through 4.5.2.9. Failure of any test shall be cause for rejection. After testing, each nozzle shall be inspected in accordance with 4.1.2.

4.4 Quality conformance inspection.

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4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105. Sampling size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination. Each nozzle shall be examined as specified in 4.5.1. The presence of one or more defects shall be cause for rejection.

4.4.3 Test.

4.4.3.1 Individual. Each nozzle shall be tested as specified in 4.5.2.3. Failure of the test shall be cause for rejection.

4.4.3.2 Sample. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1, 4.5.2.2, and 4.5.2.4 through 4.5.2.6. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The nozzle shall be examined as specified herein for the following defects:

101. Materials not as specified.
102. Materials not resistant to corrosion and deterioration, or treated to be made resistant to corrosion and deterioration in the applicable storage and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes, or treatment.
105. Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the nozzle.
106. Nozzles not finished as specified.
107. Any part missing or not of specified type or size.
108. Any part fractured, split, punctured, sprung, or malformed.
109. Any sharp burr or sliver that could be injurious to personnel.
110. Any component that is inoperative or will not function as specified.
111. Threads not as specified.
112. Parts not as specified for interchangeability.
113. Dimensions not as specified.
114. Weight not as specified.
115. Marking missing, incomplete, or not permanent.
116. Maintainability not as specified.
117. Lubrication not as specified.
118. Workmanship not as specified.
119. Human factors engineering not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified herein, tests shall be performed in ambient temperatures ranging from -60 to +95 °F.

4.5.2.2 Hydrostatic strength. With the valve seat in the closed position, the nozzle shall be subjected to an internal hydrostatic pressure employing

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MIL-C-7024, type II, Stoddard's solvent. The pressure shall be raised slowly to 150 psi in 30-psi increments. Each pressure shall be held for 1 minute. Nonconformance to 3.5.1 shall constitute failure of this test. The leakage test of 4.5.2.3 shall be performed after the test.

4.5.2.2.1 Valve body hydrostatic test. The outlet of the nozzle shall be blocked and the entire valve body subjected to a hydrostatic pressure of not less than 100 psi. Pressure shall be maintained for 1 minute. Any evidence of leaking or cracking shall constitute failure of this test.

4.5.2.3 Leakage with valve closed and outlet open. With the valve in the closed position, the outlet open and the nozzle immersed in water, the nozzle poppet shall be subjected to 2 psig air pressure applied at the nozzle inlet. The air pressure shall then be increased to 30 psig and the nozzle checked for body and poppet leakage. Pressure shall be maintained for 2 minutes for quality conformance testing and 30 minutes for first article inspection. Nonconformance to 3.5.2 shall constitute failure of this test.

4.5.2.3.1 Interface leakage. The adapter as connected to the closed-circuit refueling (CCR) nozzle shall be pressurized with MIL-C-7024, type II, Stoddard's solvent to a pressure of 30 psi. Pressure shall be maintained for 1 minute. There shall be no evidence of leakage at the interface surface between the poppet and the CCR nozzle. Nonconformance to 3.5.2 shall constitute failure of this test.

4.5.2.4 Mechanical strength. The nozzle shall be tested for mechanical strength in accordance with UL 842. With the nozzle connected to a coupler, apply a 100 foot-pound bending load to act at the lock-lug plane on the nozzle inlet adapter. Leakage test 4.5.2.3 shall be repeated after this test. Nonconformance to 3.5.3 shall constitute failure of this test.

4.5.2.5 Flow capacity. The nozzle shall be subjected to a fluid flow test with an inlet pressure of 15 ±0.5 psi. The inlet pressure shall be measured at a distance of 10 pipe diameters upstream of the nozzle inlet. Not less than 500 gallons shall be discharged through the nozzle in determining the flow in gallons per minute (see 4.5.2.7).

4.5.2.5.1 Test fluid. Except as specified herein, all tests shall be conducted using a test fluid conforming to MIL-T-5624, grade JP-4; MIL-T-83133, grade JP-8; MIL-C-7024, type II, Stoddard solvent. The test fluid temperature shall be between -60 and +95 °F. All flow rate performance data shall be corrected to a specific gravity of 0.78. Nonconformance to 3.5.4 shall constitute failure of this test.

4.5.2.6 Chain and spring test. The dust cap, chain, and spring assembly shall be subjected to a pull test of not less than 70 pounds. Nonconformance to 3.4.7 shall constitute failure of this test.

4.5.2.7 Performance. The nozzle shall be installed in a test set-up which will provide a regulated inlet pressure of 15 ±2 psi under all flow conditions from no flow to the maximum flow rate. The nozzle valve shall be cycled open and closed at a rate not faster than 6 cycles per minute. The test cycles shall include 500 cycles at each of the following temperatures: -60, -55, -50, -40, -30, -10, +30, +50, +70, and +95 °F. The test shall include an equal number of

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cycles at maximum (50 gpm), intermediate (30 gpm), and low (approximately 8 gpm) flow rates. No maintenance shall be performed on the nozzles during this test. Any evidence of external leakage, cracking, sticking of the valve, or operating handle becoming inoperative, shall constitute failure of this test.

4.5.2.8 Fuel resistance and extreme temperature test. The nozzle shall be tested as specified in table I.

4.5.2.8.1 Low temperature. The nozzle shall be tested as specified in table I and in accordance with MIL-STD-810, method 502.2, procedures I (storage), II (operation), and III (manipulation). Nonconformance to 3.4.3 and 3.10 shall constitute failure of this test.

4.5.2.8.2 High temperature. The nozzle shall be tested as specified in table I and in accordance with MIL-STD-810, method 501.2, procedures I (storage) and II (operation). Nonconformance to 3.10 shall constitute failure of this test.

4.5.2.9 Accelerated corrosion. The nozzle shall be immersed in a solution of 3.5 percent by weight NaCl in distilled water. After immersion, the solution shall be drained and the nozzle shall be heated in an oven to a temperature of 130 ±5 °F for a period of not less than 1 hour. The immersion and heating cycle shall be repeated 50 times. The nozzle shall not be operated at any time during the immersion cycles. Immediately after completing the immersion cycles the nozzle shall be washed out with warm water to remove salt accumulation. The nozzle shall be dried; wet with fuel; and tested as specified in 4.5.2.2, 4.5.2.3, and 4.5.2.4. Corrosion of any part of the nozzle to a degree that affects performance shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection.

4.6.1.1 Unit of product. The unit of product shall be a quantity of nozzles preserved and packaged in a single container.

4.6.1.2 Sampling. Sampling shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

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TABLE I. Fuel resistance and extreme temperature test schedule.

Test period 1/	Phase I, soak	Phase I, dry	Phase II, soak	Phase II, dry	Low temperature
Nozzle configuration	2/	Drained and blown dry, inlet and outlet ports open	2/	Drained and blown dry, inlet and outlet ports open	2/
Test fluid	ASTM D 471, reference fuel B	None	ASTM D 471, reference fuel B	None	ASTM D 471, reference fuel B
Period duration	96 hours minimum	24 hours	18 hours minimum	30 hours	18 hours
Ambient and test fluid	95 °F	Circulating air at 95 °F	95 °F	Circulating air at 95 °F	Lower the fluid temperature to -65 °F maximum for a minimum of 18 hours 3/
Operation or test during period	Actuate component once every 6 hours in a normal manner	None	Actuate component once every 6 hours in a normal manner	None	After 18 hours with temperature not higher than -60 °F, conduct performance (4.5.2.7) and leakage test (4.5.2.3) using ASTM D 471, reference fuel B 4/
Operation or tests immediately after period	Conduct leakage test (4.5.2.3) using ASTM D 471, reference fuel B	Actuate components for 5 cycles. Conduct leakage test (4.5.2.3) using ASTM D 471, reference fuel A	Same as phase I, soak	Same as phase I, dry	With temperature not higher than +95 °F, conduct performance (4.5.2.7) and leakage test (4.5.2.3) using ASTM D 471, reference fuel B 4/

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1/ Each period shall follow immediately after the preceding one in the order noted.
2/ The nozzle shall be maintained in such a manner as to insure complete contact of all nonmetallic parts with the fluid as would be expected under normal service conditions.
3/ Unless an increased test period is specified by the contracting officer (see 6.2).
4/ Return the nozzle to room temperature and conduct leakage test 4.5.2.3.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects:

120. Nozzles not preserved as specified for level A or C (see 5.1.1 and 5.1.2).
121. Shipping containers not as specified for levels A and C (see 5.2.1 and 5.2.2).
122. Shipping container weight and size exceeding specified limitations for levels A and C (see 5.2.1 and 5.2.2).
123. Strapping not as specified for level A (see 5.2.1).
124. Marking incorrect, missing or illegible (see 5.3).

4.7 Inspection comparison. The Government may select nozzles at any time during the contract production period and subject these nozzles to the examination specified in 4.5.1 and the test specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected first article nozzles. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional test shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of any nozzle to meet any requirement specified herein, during, and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the nozzle and shall be cause for refusal by the Government to continue acceptance of production nozzles until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on nozzles produced and accepted under the contract. Any deficiencies found as a result of first article inspection will be considered prima facie evidence that all nozzles accepted prior to completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each nozzle shall be preserved in accordance with the method III requirements of MIL-P-116, utilizing a fiberboard carton as the unit

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container; the carton shall conform with PPP-B-636, class weather resistant, grade W6c. Cushioning material conforming to PPP-C-1752 or PPP-C-1842 shall be used, as necessary.

5.1.2 Level C. Each nozzle shall be preserved as specified for level A, except the boxes shall be class domestic.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. Nozzles, preserved as specified in 5.1, shall be packed in boxes conforming with PPP-B-601, overseas type, style optional. The gross weight of the shipping container and contents shall not exceed 150 pounds. Boxes shall be strapped in accordance with the appendix to the box specification except strapping shall be flat steel and shall be finish B.

5.2.2 Level C. Nozzles, preserved as specified in 5.1, shall be packed as specified for level A, except the boxes shall be domestic type and the strapping may be finish A.

5.3 Marking. Marking of both unit and shipping containers shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The nozzle covered by this specification is intended for adapting closed-circuit refueling nozzles to conventional open port refueling in a low temperature (-60 °F) environment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is not required (see 3.2 and 4.3).
d. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
e. Level of preservation and packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that

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bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Subject term (key word) listing.

Arctic
Dust Cap
Flow control
Fuels, liquid hydrocarbon
Nozzle
Spout
Threads
Valve stem
Valves

6.5 Figure 1. Figure 1 is for informational purposes only.

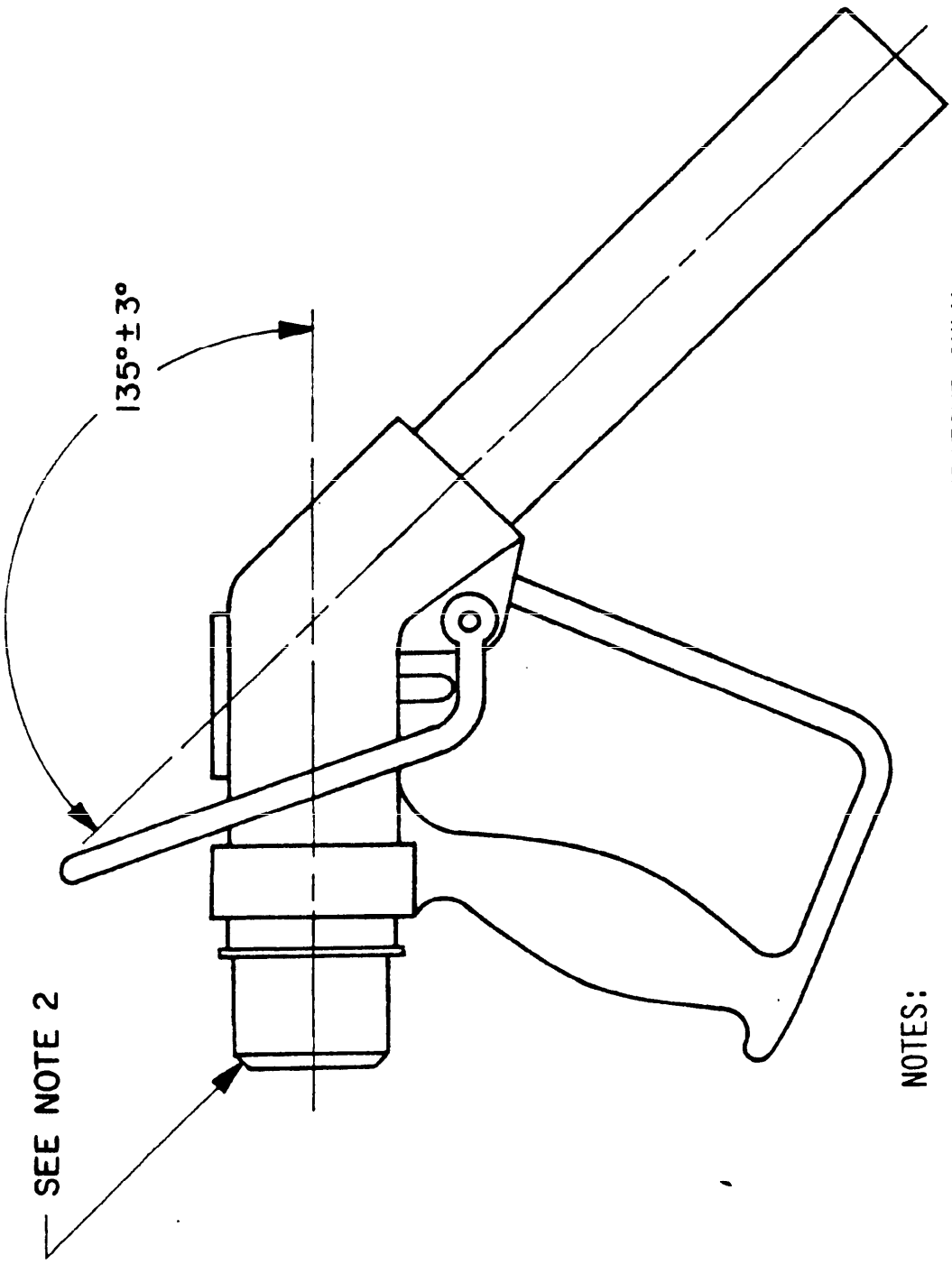
Custodian:
Army - ME

Preparing activity:
Army - ME

Project 4930-A275

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X-4857

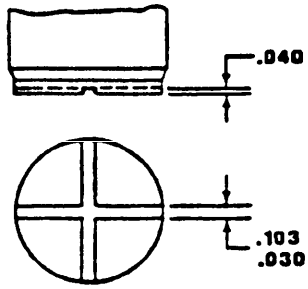


NOTES:

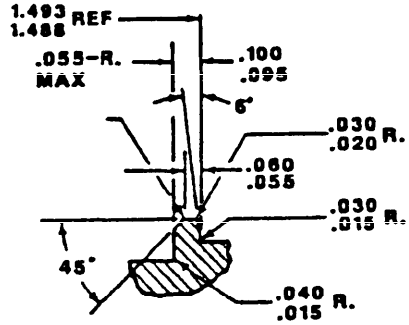
1. NOZZLE CONFIGURATION IS ILLUSTRATIVE ONLY.
2. SEE FIGURE 2 FOR INLET FITTING DIMENSION.

FIGURE 1. Adapter nozzle.

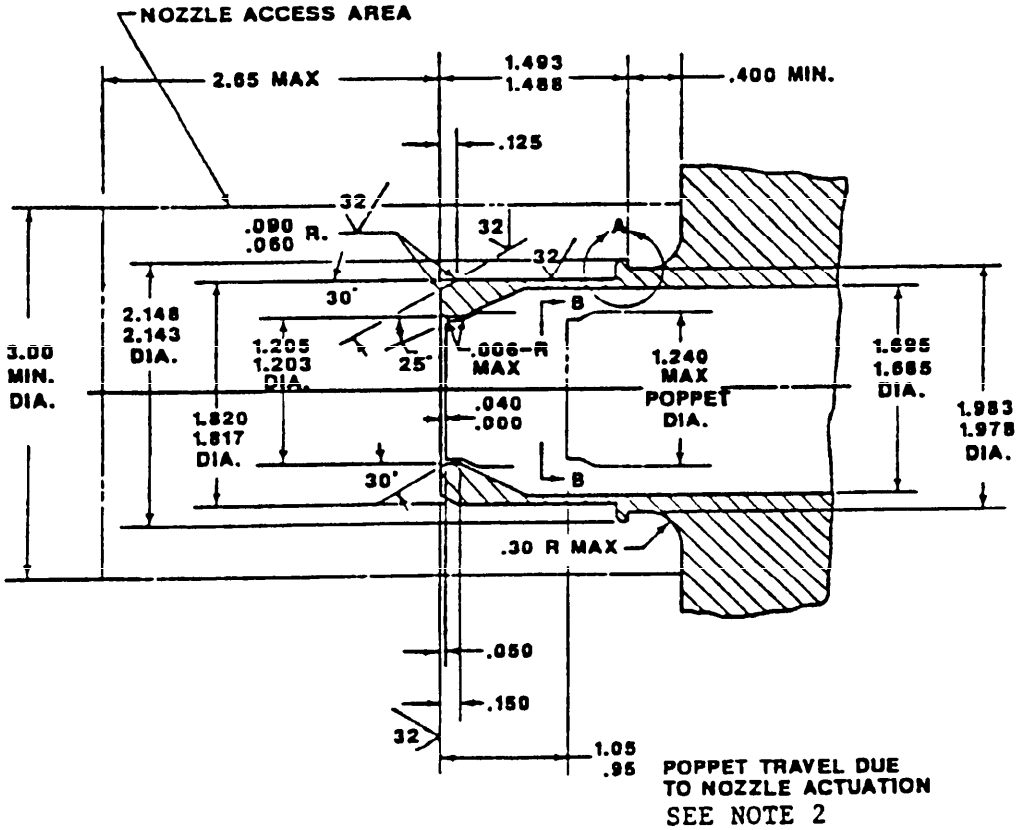
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VIEW B-B



VIEW A



NOTES:

1. TOLERANCES, UNLESS OTHERWISE SPECIFIED: DECIMALS ±.010, ANGLES ±0°30'.
2. POPPET SHALL BE SELF-SEALING WITH A SPRING LOAD OF 3 ±.3 LB SEATED, 3 LB/INCH MAXIMUM SPRING RATE.

FIGURE 2. Adapter configuration, closed-circuit fuel servicing.

X-4840

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-N-53093(ME)	90/02/07

3. DOCUMENT TITLE
Nozzle, Adapter, Closed-Circuit To Gravity Fill, Arctic Service

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717
ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE Ft. Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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